

Reference excerpt 01-001

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arXiv:1604.05171v1 [physics.med-ph] 18 Apr 2016 notes series Physical description of the blood flow from the internal jugular vein to the right atrium of the heart: new ultrasound application perspectives Francesco Sisini Department of Physics and Earth Sciences, University of Ferrara, Via Saragat 1, 44122 Ferrara, Italy Published on 5 March 2016. Ikken hissatsu () means something like to annihilate at one blow . This document is part of a series of notes each one targeting a single goal. Each note has to annihilate at one blow ! CONDITIONS OF USE This is a self-published methodological note distributed under the Creative Commons Attribution License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The note contains an original reasoning of mine and the goal to share thoughts and methodologies, not results. Therefore before using the contents of these notes, everyone is invited to verify the accuracy of the assumptions and conclusions. BACKGROUND The brain drainage is due to the venous blood flow directed from the brain to the heart through the internal jugular veins (IJVs), epidural veins and vertebral veins (see Fig. 1). The brain-heart direction indicates that there must be a negative pressure gradient driving the blood flow (i.e. the pressure in the brain is higher than the pressure in the pressure in the right atrium (RA)). The pressure, where the IJVs begin, is the residual arterial pressure and, despite its pulsatile nature, in this study it is considered to be constant.